week-2.py

```

'''
Algebra week two
'''

'''
Example problem notation:
'''

3(x - 2) + 4  =  19

#step one: distribute the 3 to the (x-2)
3x - 6 + 4  =  19

#step two: combine constants
#-6 + 4 =  -2
3x - 2  =  19

#step three: add 2 to both sides
(3x - 2) + 2  =  (19) + 2
#-2 + 2 = 0; 19 + 2 = 21
(3x)  =  (21)

#step four: divide both sides by 3
(3x) / 3  =  (21) / 3
#3 / 3 = 1; 21 / 3 = 7
x  =  7

'''
problem 1
'''

f(x) = -3x^3 - 2
g(x) = 3x + 2

Find f(-4) and g(-3)

#step one: substitute -4 for x in f(x)
f(-4) = -3(-4)^3 - 2
      = -3(-64) - 2
      = 192 - 2
      = 190

#step two: substitute -3 for x in g(x)
g(-3) = 3(-3) + 2
      = -9 + 2
      = -7

```

```
#step three: final answers
f(-4) = 190
g(-3) = -7

'''
Solving a linear equation with several occurrences of the variable: Variables on both sides and distribution
'''

7x - 15  =  3(x - 1)

#step one: distribute the 3 to the (x-1)
7x - 15  =  3x - 3

#step two: subtract 3x from both sides
(7x - 15) - 3x  =  (3x - 3) - 3x
#7x - 3x = 4x; 3x - 3x = 0
4x - 15  =  -3

#step three: add 15 to both sides
(4x - 15) + 15  =  (-3) + 15
#-15 + 15 = 0; -3 + 15 = 12
4x  =  12

#step four: divide both sides by 4
(4x / 4)  =  (12 / 4)
#4 / 4 = 1;  12 / 4 = 3
1x  =  3

'''
Graph the following functions
'''

f(x) = x
f(x) = (x^2)
f(x) = |x|
f(x) = (x^3)

f(x) = x
#step one: create a table of values for both: x and f(x)
x  | f(x)
-2 | -2
-1 | -1
 0 |  0
 1 |  1
 2 |  2

'''
graph is done in desmos; it's a straight line through the (0, 0) origin with a slope of rise: 1 / run: 1
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```
'''
f(x) = (x^2)
#step one: create a table of values for both: x and f(x)
#f(x) = each x value is squared
x  | f(x)
-2 |  4
-1 |  1
 0 |  0
 1 |  1
 2 |  4

'''
graph is done in desmos; it's a skinny parabola opening **upwards** with the vertex (center) at the (0, 0) origin
why is the parabola skinny? because the coefficient of x^2 is 1, which is a small number; if it were a larger number, the parabola would be wider
so if the graph was f(x) = 10x^2, the parabola would be fatter than f(x) = x^2
'''

f(x) = |x|

#step one: create a table of values for both: x and f(x)
x  | f(x)
-2 |  2
-1 |  1
 0 |  0
 1 |  1
 2 |  2

'''
graph is done in desmos; it's a v-shaped graph with the vertex (center) at the (0, 0) origin

why is it v-shaped? because the absolute value of a number is always positive, so the graph will always be above the x-axis

so the line will be on the left if the y-axis is negative, and on the right if the y-axis is positive
but it will never be on the bottom since the x-axis will always be positive
'''

'''
Calculate the distance between the points in the coordinate plane
'''

E = (0,7)  and  M = (6,-1)

#step one: use the distance formula to find the distance between the two points

distance = √‾((x₂ - x₁)² + (y₂ - y₁)²)

distance = √‾((6 - 0)² + (-1 - 7)²)
```

```
           = √‾‾(6² + -8²)
           = √‾‾(36 + 64)
           = √‾‾100
           = 10

'''
Calculate the distance between the two points
'''

G = (5,-9)  and  F = (8,-1)

#step one: use the distance formula

distance = √‾‾((x₂ - x₁)² + (y₂ - y₁)²)

distance = √‾‾((8 - 5)²  + (-1 - (-9))²)
           = √‾‾(3² + (-1 + 9)²)
           = √‾‾(9 + (8)²)
           = √‾‾9 + 64
           = √‾‾73

'''
Calculate the distance between the two points
'''

L = (-7,-8)  and  H = (1,-1)

#step one: use the distance formula

distance = √‾‾((x₂ - x₁)² + (y₂ - y₁)²)

distance = √‾‾((1 - (-7))² + (-1 - (-8))²)
           = √‾‾((1 + 7)² + (-1 + 8)²)
           = √‾‾(64 + 49)
           = √‾‾(113)

'''
Evaluate
'''
-(-4)³  and  - (7)²


-(-4)³
-(-64)
64

-(7)²
-(49)
```

```
'''
Solve for v; simplify your answer
'''
-4(v + 3)  =  4v + 28


#step one: distribute -4 on left
(-4v) + (-12)  =  4v + 28


#step two: add 4v to both sides
(4v) + (-4v) + (-12)  =  (4v) + 4v + 28


#step three: combine like terms
0 + (-12)  =  8v + 28


#step four: divide both sides by 8
(-12) / 8  =  ((8v) / 8) + (28 / 8)


#step five: reduce terms
(-3)/2  =  1v + (7/2)


#step six: subtract (7/2) from both sides
((-3)/2) + ((-7)/2)  =  (1v + (7/2)) + ((-7)/2)

((-10)/2)  =  1v


#reduce
(-5)/1  =  1v
(-5)  =  v


'''
simplify
'''
((v)⁵ (u)⁷)  /  ((v)⁴ (u)¹)

#v's expnts: 5 - 4 = 1
#u's expnts: 7 - 1 = 6
#this expnt subtraction removes the denominator

(v)¹ (u)⁶
v (u)⁶

'''
```

```
244 simplify
245 '''
246 ((x)⁶ (z)⁴)  /  (x)⁶ (z)⁵
247 
248 #x's expnts: 6 - 6 = 0
249 #z's expnts: 4 - 5 = (-1)
250 
251 (x)⁰ (z)⁻¹
252 
253 #(x)⁰ = 1
254 #(z)⁻¹ = (1/z)
255 
256 1 (1 / z)
257 1/z
258 
259 '''
260 simplify
261 '''
262 
263 ((a)⁴ (b)²)  /  ((a)³ (b)⁶)
264 
265 #(a) 4 - 3 = 1
266 #(b) 2 - 6 = (-4)
267 
268 (a)¹ (b)⁻⁴
269 
270 #(a) = (a) / 1
271 #(b)⁻⁴ = (1/b⁴)
272 ((a) / 1) (1 / b⁴)
273 
274 #neither the 1 in the denominator of (a), nor the 1 in the numerator of (b⁴) is necessary. so we remove them and then we're left with (a) in the numerator and (b⁴) in the denominator
275 (a) / b⁴
276 
277 '''
278 simplify
279 '''
280 ((b)⁶ (a)³)  /  ((b)⁷ (a))
281 
282 #(b) 6 - 7 = -1
283 #(a) 3 - 1 = 2
284 
285 (b)⁻¹ (a)²
286 
287 #(b)⁻¹ = (1/b¹) = (1/b)
288 #see note in line 274
289 
290 (a)² / (b)
291 
292 '''
```

```
simplify
'''
((x)⁴ (z)⁵)  /  (3(x)⁴ (z)³)

#(x)⁴ - (x)⁴ = (x)⁰ = (1)
#(z)⁵ - (z)³ = (z)²

((1) (z)²)  /  3
(z)² / 3

'''
simplify
'''

((a)⁷ (b)⁶)  /  ((a) (b)⁴)

#(a)⁷ - (a)¹ = (a)⁶
#(b)⁶ - (b)⁴ = (b)²

(a)⁶ (b)²

'''
simplify
'''

((b)⁷ (a))  /  ((b)⁶ (a)⁴)

#(b)⁷ - (b)⁶ = (b)¹
#(a)¹ - (a)⁴ = (a)⁻³
(b)¹ (a)⁻³

#(a)⁻³ = (1 / a³)
(b) / (a³)

'''
find the value
'''
³√‾‾8

#what number = 8 after being cubed?
#(2)³ = 8
2

'''
find the value
'''
³√‾‾(-216)

#what number = -216 after being cubed?
#because the answer is negative, I need to cube a negative number
```

```
#(-6)³ = -216
-6

'''
rewrite w/out parentheses and simplify
'''
(x + 2)²

#shortcut formula --->
#square the first term +  double the product of the two terms + square the second term

#(a + b)² = a² + 2ab + b²
#a = x
#b = 2

(x)² + 2(x)(2) + (2)²
(x)² + 4(x) + 4

'''
rewrite w/out parentheses and simplify
'''
(4u - 5)²

#shortcut formula --->
#square the first term - double the product of the two terms + square the second term

#(a - b)² = a² - 2ab + b²
#a = 4u
#b = 5

(4u)² - 2(4u)(5) + (5)²
(16u²) - 2(20u) + 25
(16u²) - 40u + 25

'''
rewrite w/out parentheses and simplify
'''

(2 + 3y)²
(3y + 2)²

#shortcut formula --->
#square the first term + double the product of the two terms + square the second term

#(a + b)² = a² + 2ab + b²
#a = 3y
#b = 2

(3y)² + 2(3y)(2) + (2)²
(9y)² + 2(6y) + 4
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(9y)² + (12y) + 4

'''
simplify the square root of a whole number < 100
'''

√‾‾8

#find the largest perfect square factor ---> 4
8 = 4 * 2

#rewrite square root of 8 as square root of 4 * 2
√‾‾8  =  √‾‾(4 × 2)

#separate the square roots
√‾‾8  =  √‾‾4 * √‾‾2

#simplify √‾‾4
√‾‾8  =  2 * √‾‾2
2 * √‾‾2

'''
simplify the square root of a whole number < 100
'''
√‾‾27

#find the largest perfect square factor ---> 9
27 = 9 * 3

#rewrite square root of 27 as square root of (9 * 3)

√‾‾27  =  √‾‾(9 * 3)

#separate the square roots
√‾‾27  =  √‾‾9 * √‾‾3

#simplify √‾‾9
√‾‾27  =  3 * √‾‾3
3 * √‾‾3


'''
simplify the square root of a whole number < 100
'''
√‾‾32

#find the largest perfect square factor ---> 16
32 = 16 * 2

#rewrite square root of 32 as square root of (16 * 2)
```

```
√‾‾32  =  √‾‾(16 * 2)

#separate the square roots
√‾‾32  =  √‾‾16 * √‾‾2

#simplify √‾‾16
√‾‾32  =  4 * √‾‾2
4 * √‾‾2
```